Alumni Management System for TU

A report submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology
in
Computer Science and Engineering
Of
Tezpur University

Submitted By-

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Declaration by the Student

We hereby declare that the project work presented in this report entitled "ALUMNI MANAGEMENT SYSTEM FOR TU", submitted in partial fulfilment for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** during the academic year 2024-2025, has been carried out by us and that it has not been submitted in part or whole to any institution for the award of any other degree or diploma.

Date: 27th May 2024 ANIKET ROY

Place: TEZPUR UNIVERSITY

Department of Computer Science and Engineering

Tezpur University

Tezpur-784028, Assam, India

CERTIFICATE

This is to certify that the report entitled in "Alumni Management System for TU" is a bona fide record of project done by Aniket Roy (CSB21028) and submitted in fulfillment of the requirements and the regulations of the mini project of 3rd year of the 4-year course of Bachelor of Technology in Computer Science and Engineering, Tezpur University. He is carrying out the project work under the supervision of Dr. Rajiv Goswami, Tezpur University.

This certification does not necessarily endorse or accept every statement made, opinion expressed or conclusion drawn as recorded in the report. It only signifies the acceptance of this report for the purpose for which it is submitted.

Date:15 May 2024 Dr. Sarat Saharia

Place: Tezpur University HOD

ACKNOWLEDGEMENT

Apart from the efforts of myself, I take this opportunity to express my gratitude to the people who have been instrumental in the successful completion of this project. would like to show my greatest appreciation to my Guide **Dr. Rajiv Goswami** sir. I can't say thank you enough for his tremendous support and help. Without his encouragement and guidance this project would not have materialized.

INTRODUCTION

The Alumni Management System of Tezpur University is a platform designed to nurture the community of alumni, fostering lifelong engagement with the university. It combines technology and tradition, facilitating seamless interaction among alumni, students, faculty, and administration. The system's user-friendly interface streamlines administrative tasks and fosters a sense of belonging within the university family. It uses data analytics and automation to enable personalized communication, targeted outreach, and strategic decision-making, enhancing the alumni experience and enriching the university's institutional memory.

The Alumni Management System is a testament to Tezpur University's legacy and progressive vision, propelling the institution towards greater excellence and impact in the ever-evolving higher education landscape.

PROBLEM STATEMENT

The lack of an efficient Management System for Alumni's of our university poses challenges in maintaining robust engagement and effective communication. Existing methods are outdated, leading to poor data management, limited interaction with alumni, and missed opportunities for collaboration and support. Without a streamlined system, we struggle to harness the potential of alumni networks for fundraising, career guidance, and academic enrichment.

My project aims to develop a comprehensive Alumni Management System that facilitates seamless communication, data tracking, and alumni involvement, empowering our university to foster lifelong relationships, enhance institutional reputation, and leverage alumni expertise for mutual benefit.

FEASIBILITY STUDY

1. Technical Feasibility

The Technical Feasibility of implementing an Alumni Management System relies on several factors:

- **Software Infrastructure:** Assessing the current technological infrastructure of the university to ensure compatibility with the proposed system. Compatibility issues may arise due to varying hardware specifications, operating systems, and network configurations.

- **Scalability:** The system must be capable of handling a large volume of data, including alumni profiles, contact information, academic records, and engagement activities. Scalability ensures smooth operation as the alumni network grows over time.

- **Integration:** Integration with existing systems such as student databases, CRM software, and communication platforms is essential for seamless data exchange and functionality. Compatibility with APIs and standardized data formats is crucial for successful integration.

- **Security:** Implementing robust security measures to protect sensitive alumni data from unauthorized access, data breaches, and cyber threats. This includes encryption protocols, access controls, and regular security audits to ensure compliance with data protection regulations.

- **Maintenance and Support:** Evaluating the availability of technical expertise within the university or outsourcing options for system maintenance, updates, and troubleshooting. Adequate technical support ensures the system's continuous operation and performance optimization. |

2. Economic Feasibility

The Economic Feasibility of the Alumni Management System involves analysing the financial aspects associated with its development, implementation, and long-term maintenance:

- **Cost-Benefit Analysis:** Conducting a comprehensive cost-benefit analysis to evaluate the projected expenses against the anticipated benefits.
  This includes one-time expenses such as software development, hardware acquisition, and implementation costs, as well as recurring expenses like maintenance, support, and personnel training.

- **Return on Investment (ROI):** Estimating the potential ROI from improved alumni engagement, fundraising efforts, enhanced reputation, and operational efficiencies facilitated by the system. Tangible benefits such as increased donations, reduced administrative overhead, and enhanced alumni satisfaction contribute to the overall economic feasibility.

- **Budget Allocation:** Allocating sufficient budgetary resources for the development, implementation, and ongoing maintenance of the Alumni Management System. Prioritizing expenditures based on critical requirements and expected outcomes ensures prudent financial management and maximizes the system's economic viability.

- **Funding Sources:** Exploring funding sources such as grants, sponsorships, alumni donations, and institutional budgets to finance the project. Collaboration with alumni associations, corporate partners, and government agencies can provide additional financial support and enhance the project's economic feasibility.

Operational Feasibility

The Operational Feasibility of the Alumni Management System assesses its practicality and suitability for day-to-day operations within the university:

- **User Acceptance:** Evaluating the willingness and readiness of stakeholders, including university staff, alumni volunteers, and administrators, to adopt and utilize the system. User training, education, and change management initiatives are essential for fostering acceptance and minimizing resistance to system implementation.

- **Workflow Integration:** Analysing how the Alumni Management System aligns with existing operational workflows and processes within the university. Seamless integration ensures minimal disruption to daily activities and enhances user productivity and efficiency.

- **Resource Availability:** Assessing the availability of human resources, infrastructure, and administrative support necessary for the successful deployment and utilization of the system. Adequate staffing, training programs, and infrastructure upgrades may be required to support system operations effectively.

- **Customization and Flexibility:** Designing the system with customizable features and flexible configurations to accommodate diverse user needs, institutional requirements, and evolving alumni engagement strategies. Tailoring the system to specific user preferences enhances its operational feasibility and user satisfaction.

SRS DOCUMENT

1. Introduction

i) Purpose: The Alumni Management System (AMS) is designed to streamline and enhance communication and engagement between an educational institution and its alumni. It serves as a comprehensive platform to maintain a strong connection with former students, leverage their expertise, and foster a sense of belonging within the alumni community.

ii) Scope*:* The Alumni Management System manages alumni information, provides communication channels, organizes events, offers career services, and generates insights through data analysis. It also supports alumni with career guidance, job postings, mentorship programs, and networking opportunities. Its functionality helps institutions effectively manage alumni relations.

3. Overall Description

i. Product Perspective:

The Alumni Management System is a comprehensive platform designed to facilitate interaction and engagement between alumni and their alma mater. It serves as a standalone system, accessible through both web and mobile interfaces. With its user-friendly design, the system aims to bridge the gap between alumni and the educational institution, fostering a vibrant community and supporting lifelong learning.

ii. User Classes and Characteristics:

The system is built around various user classes, each with distinct characteristics:

Alumni:

- The primary users of the system, alumni can create and update their profiles, search for and connect with other alumni, participate in events, and contribute to discussions.
- They can also share their professional experiences and achievements, fostering a vibrant and engaged community.

Administrators:

- Administrators manage the system, overseeing user verification, content moderation, and system maintenance.
- They can also organize events, send out newsletters, and generate reports on alumni engagement and activities.

Faculty/Staff:

- Faculty and staff members can interact with alumni, share updates about the institution, and participate in discussions and events.
- They play a crucial role in maintaining the connection between the institution and its alumni.

Students:

- Current students can access the system to connect with alumni for mentorship, internships, and job opportunities.
- They can learn from alumni experiences and start building their network early.

Guests:

- Guests, such as potential students or employers, can access public sections of the system to learn more about the institution and its alumni.

iii. Operating Environment:

The Alumni Management System operates seamlessly across web and mobile environments. It is compatible with the latest versions of major web browsers such as Chrome, Firefox, Safari, and Edge, ensuring accessibility for users regardless of their preferred platform. Additionally, the system supports popular mobile operating systems, including iOS and Android, enabling on-the-go access to its features and resources.

iv. Design and Implementation Constraints:

To ensure the security, scalability, and efficiency of the platform, it adheres to a set of design and implementation constraints:

- **Security Standards:** The system employs robust security measures to safeguard user data and transactions, adhering to industry best practices and compliance requirements.
- **Scalability Requirements:** The architecture is designed to accommodate growth in user base and content volume, ensuring optimal performance even as the platform expands.
- **Third-Party Service Integration:** Integration with third-party services and APIs enhances the platform's functionality and enriches the user experience, while also adhering to relevant privacy and data protection regulations.

3. Functional Requirements

Functional requirements describe the specific functionalities and features that the Alumni Management System (AMS) must have in order to meet the needs of its users. These may include:

- **User authentication and authorization:** Allow alumni, administrators, and other authorized users to log in securely with appropriate access levels.

- **Profile management:** Enable alumni to create and update their profiles, including personal information, education history, employment details, and contact information.
- **Alumni directory:** Provide a searchable directory of alumni profiles, allowing users to find and connect with fellow alumni based on various criteria such as graduation year, location, or industry.
- **Event management:** Facilitate the organization and promotion of alumni events such as reunions, networking sessions, and professional development workshops.
- **Communication tools:** Implement features for sending messages, notifications, and announcements to alumni individually or in groups.
- **Job board:** Offer a platform for posting job opportunities relevant to alumni and facilitating connections between alumni and potential employers.
- **Donation management:** Enable alumni to make donations to the institution or specific programs/initiatives, with tracking and reporting capabilities.

4. Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints that the AMS must adhere to, often related to performance, usability, security, and scalability. These may include:

- **Performance:** Ensure the system responds quickly to user interactions, with minimal latency and downtime.
- **Usability:** Design an intuitive user interface that is easy to navigate and accessible to users with varying levels of technical proficiency.
- **Security:** Implement robust authentication mechanisms, data encryption, and access controls to protect sensitive alumni information from unauthorized access or data breaches.
- **Scalability:** Build the system to accommodate a growing number of users and data volume over time without significant degradation in performance.
- **Reliability:** Ensure the AMS is dependable and available when needed, with built-in redundancy and failover mechanisms to minimize disruptions.

- **Compliance:** Adhere to relevant data privacy regulations (e.g., GDPR, CCPA) and institutional policies governing the handling of alumni data.

5. Goals and Objectives

The goals and objectives of the Alumni Management System outline the intended outcomes and benefits that the project aims to achieve. These may include:

- **Enhancing alumni engagement:** Foster stronger connections between alumni and the institution by providing opportunities for networking, collaboration, and mentorship.

- **Strengthening fundraising efforts:** Increase alumni participation in fundraising campaigns and donations by providing convenient online donation channels and showcasing the impact of contributions.

- **Improving institutional reputation:** Showcase the achievements and success stories of alumni to enhance the reputation and credibility of the institution.

- **Supporting lifelong learning:** Provide resources and opportunities for alumni to continue their education and professional development, thereby contributing to their ongoing success.

- **Facilitating knowledge sharing:** Enable alumni to share expertise, insights, and experiences with current students, faculty, and fellow alumni, enriching the learning and professional development ecosystem within the institution.

DESIGN

I. Software Architecture

i. DFD (Data Flow Diagram)

- **Level 0 DFD**

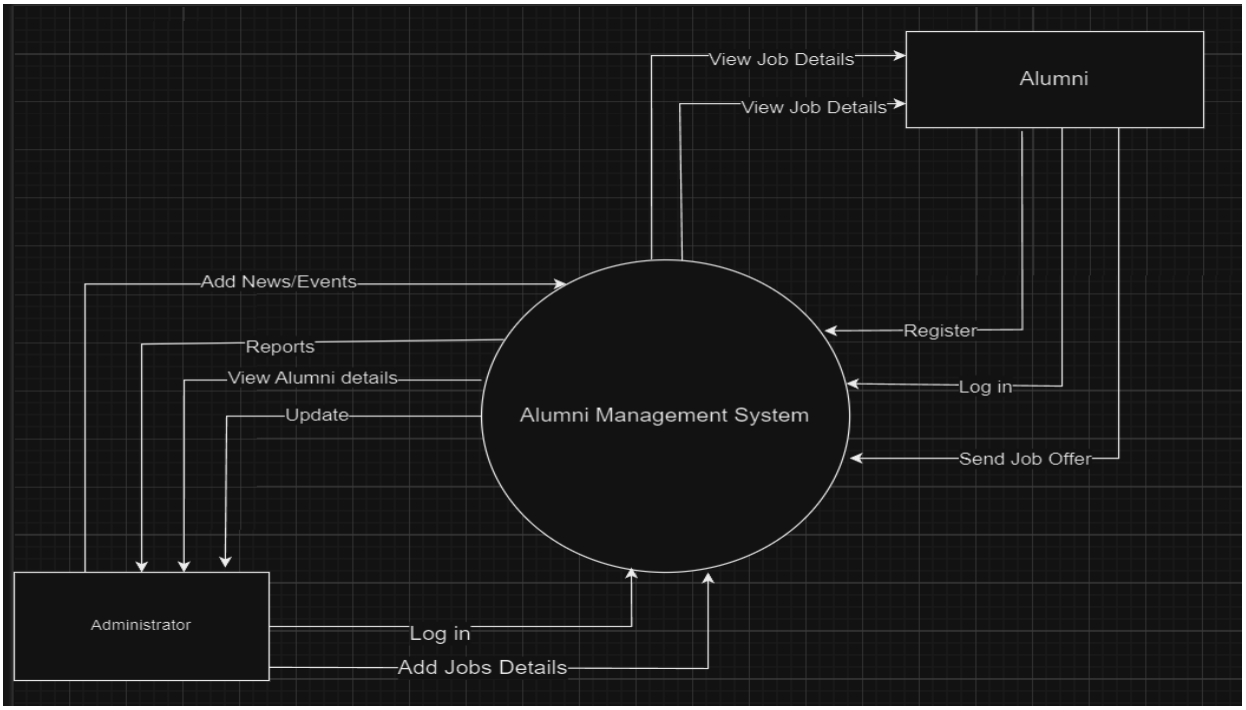


Fig: Level 0 DFD Diagram

i) External Entities:

- **Alumni:** This entity represents the former students or graduates of an educational institution who are members of the alumni network. They interact with the system to register, log in, view job details, and receive job offers.

- **Administrator:** This entity represents the administrative personnel responsible for managing the Alumni Management System. They can log in,

add news/events, generate reports, view alumni details, update information, add job details, and perform other administrative tasks.

ii) Processes:

• **Alumni Management System:** This is the central process that coordinates and manages all the functionalities related to alumni activities. It handles various operations such as user authentication, data management, job postings, event/news updates, and report generation.

• **Register:** This process allows new alumni to sign up and create an account within the system, providing their personal and professional details. • Log in: Both alumni and administrators can authenticate themselves through this process by entering their credentials, granting them access to the respective features and functionalities based on their roles.

• **Send Job Offer:** This process enables the system to send job offers or opportunities to registered alumni, facilitating their career growth and job search efforts. • View Job Details: Alumni can use this process to view comprehensive details about job openings, including job descriptions, requirements, and other relevant information.

iii) Data Stores:

• **Alumni:** This data store maintains the personal and professional information of all registered alumni members, such as their contact details, educational background, work experience, and other relevant data.

iv) Data Flows:

• **Register:** This data flow represents the transfer of alumni registration information from the alumni entity to the Alumni Management System process.

• **Log in:** Both alumni and administrators initiate this data flow by providing their login credentials to the Alumni Management System process for authentication.

- **View Job Details:** This data flow occurs when alumni request to view job details, triggering the retrieval of relevant job information from the Alumni Management System process.

- **Add News/Events:** Administrators can input news and event details through this data flow, which gets processed and stored by the Alumni Management System.

- **Reports:** Administrators can request various reports related to alumni activities, initiating this data flow from the Alumni Management System process.

- **View Alumni details:** This data flow enables administrators to retrieve and view detailed information about registered alumni from the Alumni data store.

- **Update:** Administrators can initiate this data flow to update or modify existing alumni information stored in the Alumni data store.

- **Add Jobs Details:** Administrators use this data flow to input and add new job details into the system, which are then processed and stored for future reference.

LEVEL 1 DFD (Administrator)

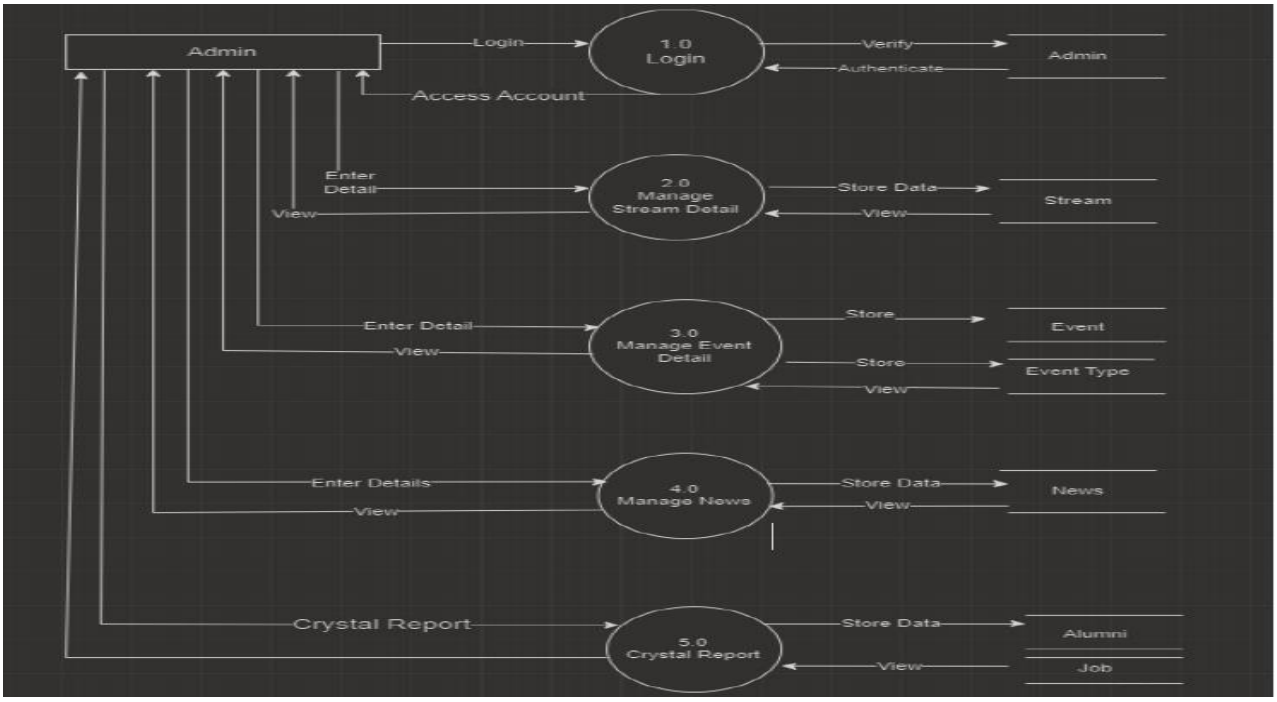


Fig: Level 1 DFD Diagram (Administrator)

i) External Entities:

- **Administrator:** This entity represents the administrative personnel responsible for managing the Alumni Management System. They interact with the system by providing their login credentials, updating portal information, adding job offers, and adding news/events.

ii) Processes:

- **Log In:** This process authenticates the administrator by verifying the provided username and password credentials against the stored account information.

- **View Admin Portal:** After successful authentication, this process retrieves and displays the administrative portal interface, allowing the administrator to access various functionalities and information.

- **Update Portal:** This process enables the administrator to modify or update the content and information displayed on the administrative portal, ensuring that it remains current and relevant.

- **Add Job Offer:** Through this process, the administrator can input and publish new job offers or opportunities for alumni members, including details such as job descriptions, requirements, and application procedures.

- **Add News/Events:** This process allows the administrator to add and share news, announcements, or upcoming events relevant to the alumni community, keeping them informed and engaged.

iii) Data Stores:

- **Account Info:** This data store maintains the login credentials (username and password) and other relevant account information for administrators, enabling secure access and authentication.

- **Portal Info:** This data store contains the information and content displayed on the administrative portal, such as alumni profiles, job listings, news, and event details.

- **Job Info:** This data store stores and manages the details of job offers and opportunities posted by the administrator for alumni members.

- **Event Info:** This data store maintains the information related to news, announcements, and upcoming events that are shared with the alumni community through the system.

iv) Data Flows:

- **Username/Password:** This data flow represents the administrator providing their login credentials (username and password) to the Log In process for authentication.

- **Account Info:** This data flow occurs when the Log In process retrieves and verifies the administrator's account information from the Account Info data store during the authentication process.

- **Portal Info:** After successful authentication, this data flow enables the View Admin Portal process to fetch and display the relevant portal information from the Portal Info data store.

- **Update:** This data flow represents the administrator updating or modifying the content and information in the Portal Info data store through the Update Portal process.

- **Job Info:** The administrator initiates this data flow by providing job offer details to the Add Job Offer process, which stores the information in the Job Info data store.

- **Event Info:** This data flow occurs when the administrator inputs news, announcements, or event details through the Add News/Events process, which are then stored in the Event Info data store.

LEVEL 1 DFD (Alumni)

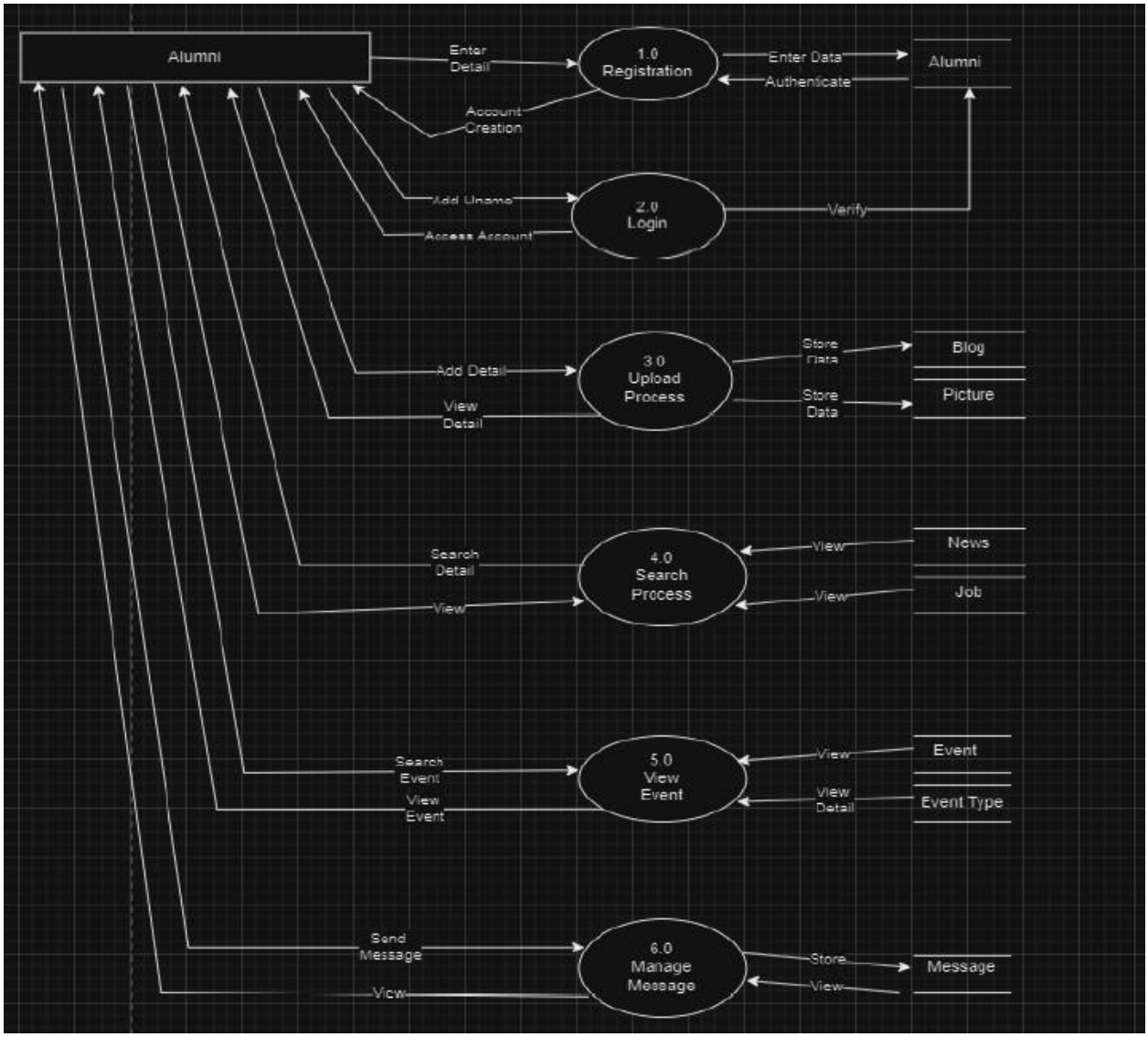


Fig: Level 1 DFD Diagram (Alumni)

i) External Entities:

- **Alumni:** This entity represents the former students or graduates who are members of the alumni community. They interact with the system by providing login

credentials, registering their information, updating profiles, and viewing job details and opportunities.

ii) Processes:

- **Log In:** This process authenticates alumni members by verifying their provided username and password against stored account information, granting them access to the system's features.
- **Register:** This process allows new alumni members to create an account by providing personal and professional details, which are then stored in the system.
- **View Profile:** After authentication, this process retrieves and displays the alumni member's personal and professional information, allowing them to review and update their profile.
- **Update Portal:** This process enables alumni members to modify or update their personal and professional information stored in the system, ensuring accurate profiles.
- **Add Job Offer:** Through this process, new job offers or opportunities are added to the system, making them available for alumni members to view and potentially apply for.
- **View Job Details:** This process retrieves and displays detailed information about job offers, such as descriptions, requirements, and application procedures, helping alumni make informed career decisions.

iii) Data Stores:

- **Account Info:** This data store maintains login credentials (username and password) and other relevant account information for alumni members, enabling secure access and authentication.
- **Alumni Info:** This data store contains personal and professional information of registered alumni members, including contact details, educational background, work experience, and other relevant data.

- **Job Info:** This data store stores and manages the details of job offers and opportunities made available to the alumni community through the system.

iv) Data Flows:

- **Username/Password:** This data flow represents alumni members providing their login credentials (username and password) to the Log In process for authentication.
- **Account Info:** This data flow occurs when the Log In process retrieves and verifies the alumni member's account information from the Account Info data store during authentication.
- **Alumni Info:** This data flow enables the Register process to store the personal and professional information provided by new alumni members in the Alumni Info data store.
- **Update:** This data flow represents alumni members updating or modifying their personal and professional information stored in the Alumni Info data store through the Update Portal process.
- **Job Info:** This data flow allows the Add Job Offer process to store and manage the details of new job offers and opportunities in the Job Info data store.
- **View Job Details:** After authentication, this data flow enables alumni members to retrieve and view the details of available job offers and opportunities from the Job Info data store.

II) STRUCTURED CHART

Main Module:

- The core module of the system responsible for coordinating between different modules and managing overall system functionality.

Authentication Module:

- Handles user authentication and access control.
- Allows alumni, administrators, and other users to log in securely.

User Registration Module:

- Facilitates the registration process for alumni. Collects and stores alumni information securely.

Alumni Profile Module:

- Manages alumni profiles.
- Allows alumni to update their information.
- Provides administrators with access to alumni profiles for management purposes.

Search and Filter Module:

- Enables users to search for specific alumni based on various criteria like graduation year, location, industry, etc.
- Implements advanced filtering options for more refined search results.

Communication Module:

- Provides features for communication between alumni, administrators, and other users.
- Supports messaging, email notifications, and alerts.

Admin Module:

- Controls administrative functionalities.

- Includes features for managing users, alumni data, and system configurations.

Dashboard Module:

- Presents a graphical interface for administrators to monitor system activity, user interactions, and data analytics.

User Management:

- Allows administrators to manage user accounts, permissions, and roles.

- Provides functionalities for adding, editing, and deleting user accounts.

Alumni Data Management:

- Handles the storage, retrieval, and manipulation of alumni data. Ensures data integrity and security.
- Communication Management: Manages communication channels and protocols within the system.
- Implements features for automated notifications, reminders, and announcements.

III) DATA DICTIONARY

1. Table: Alumni

- **AlumniID:** Unique identifier for each alumnus (Primary Key, Integer)
- **FirstName:** First name of the alumnus (String)
- **LastName:** Last name of the alumnus (String)
- **Email:** Email address of the alumnus (String)
- **GraduationYear:** The year the alumnus graduated (Integer)
- **Course:** The course the alumnus studied (String**)**
- **ContactNumber:** Contact number of the alumnus (String)
- **CurrentJob:** Current job of the alumnus (String)
- **CurrentCity:** Current city of residence (String)

2. Table: Events

- **EventID:** Unique identifier for each event (Primary Key, Integer)
- **EventName:** Name of the event (String)
- **EventDate:** Date of the event (Date)
- **EventLocation:** Location of the event (String)
- **EventDescription:** Description of the event (String)

3. Table: Registrations

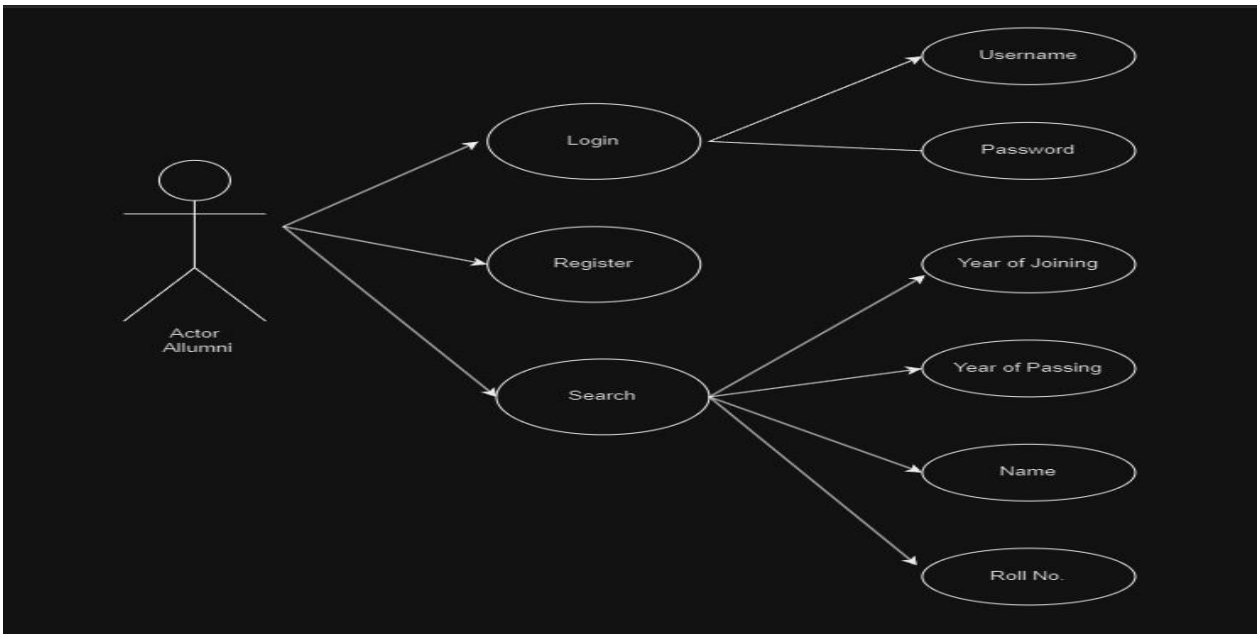
- **RegistrationID:** Unique identifier for each registration (Primary Key, Integer)

- **AlumniID:** ID of the alumnus who registered for the event (Foreign Key referencing Alumni, AlumniID, Integer)

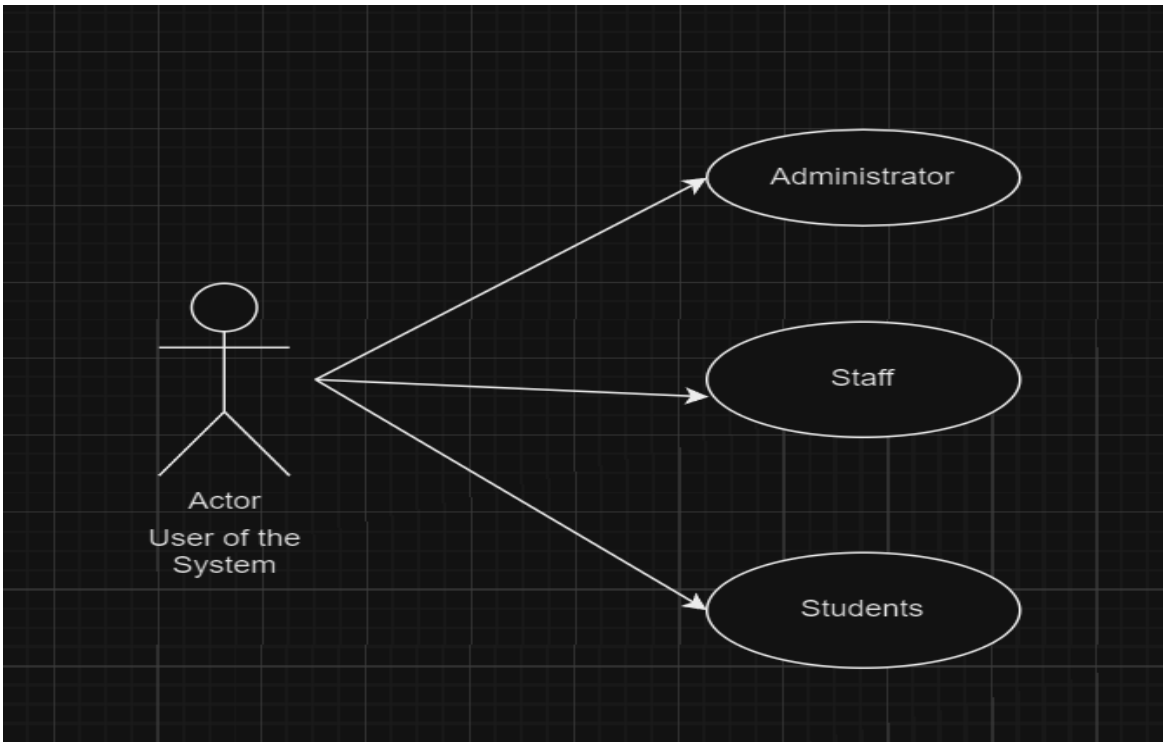
- **EventID:** ID of the event for which the alumnus registered (Foreign Key referencing Events, EventID, Integer)

- **RegistrationDate:** Date when the alumnus registered for the event (Date)

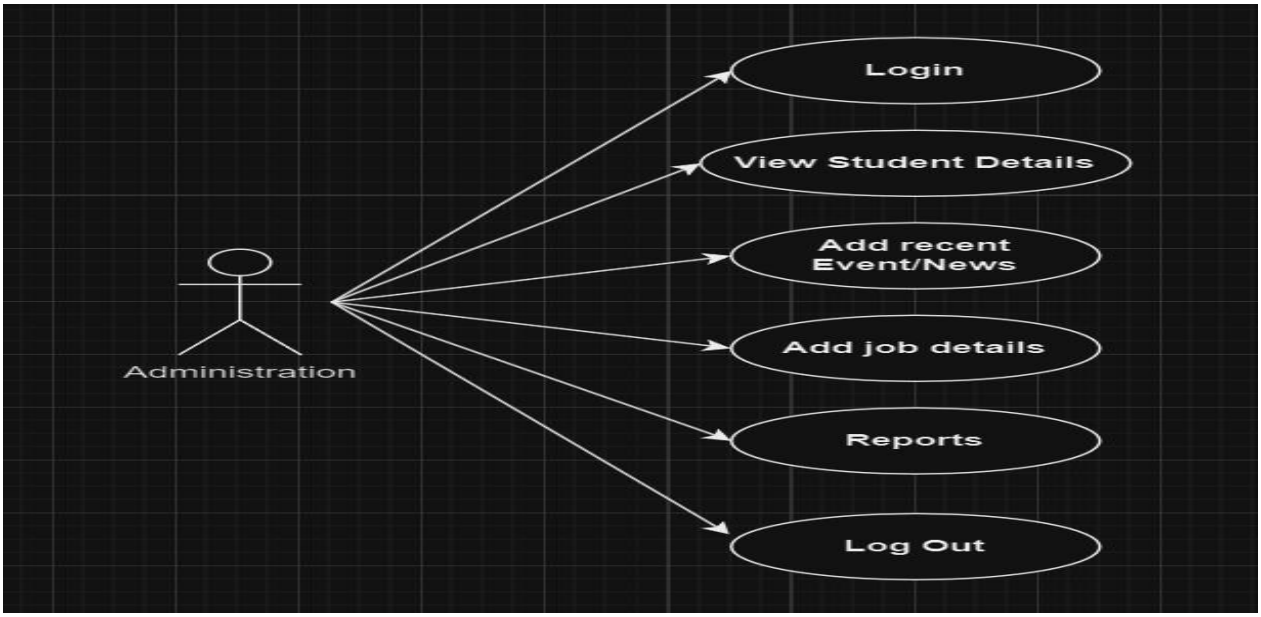
• USE CASE DIAGRAM



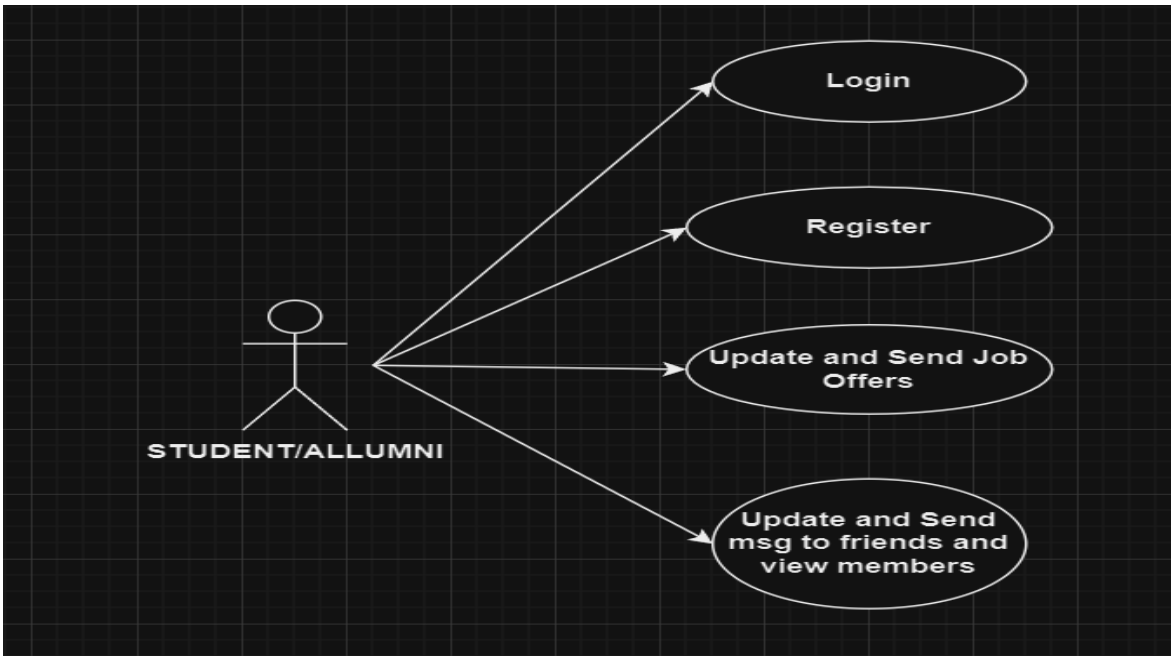
- **USER'S OF THE SYSTEM**



- **ADMINISTRATOR OF THE SYSTEM**



- **STUDENTS/ALLUMNI OF THE SYSTEM**



UML Diagram

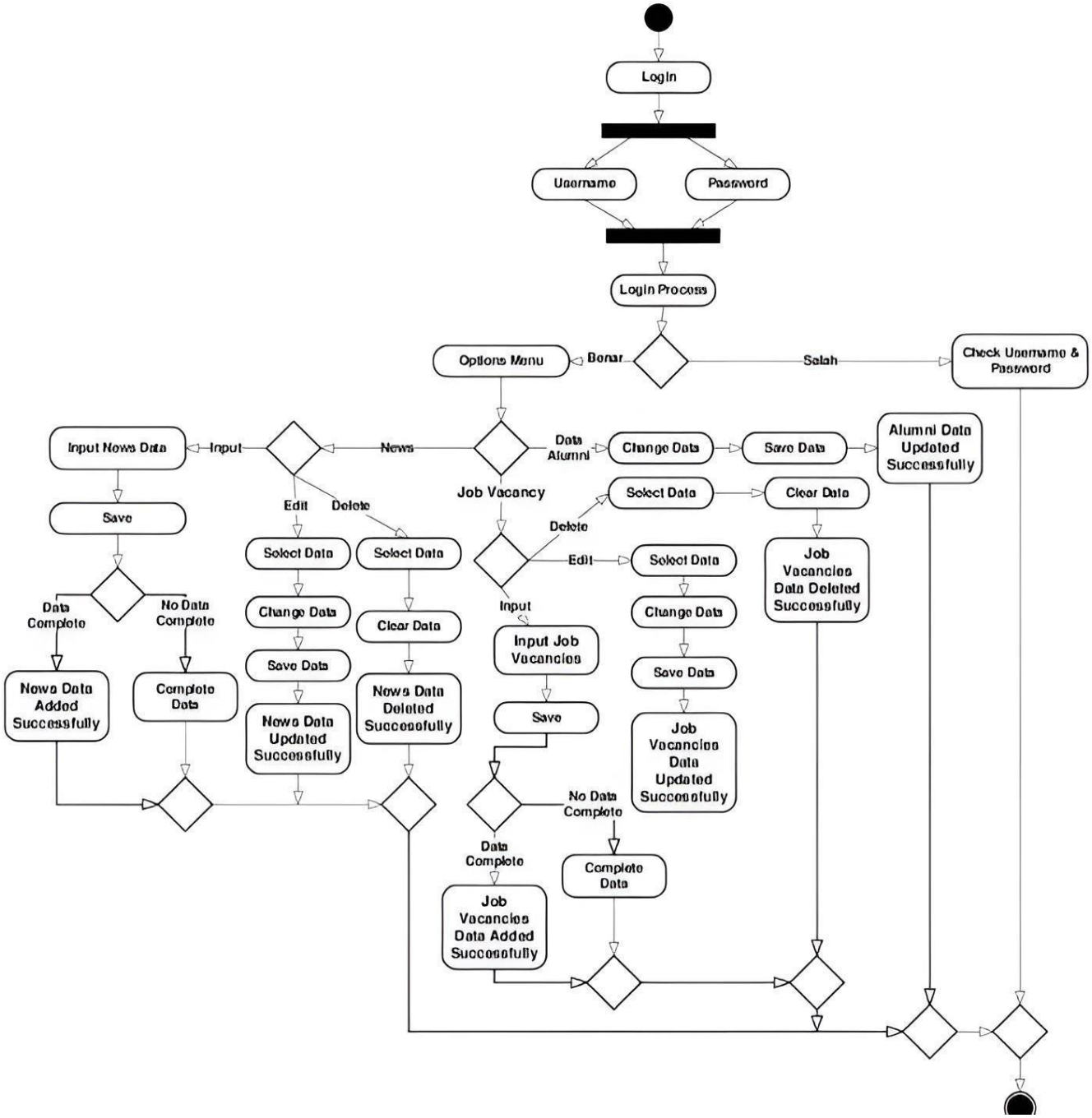
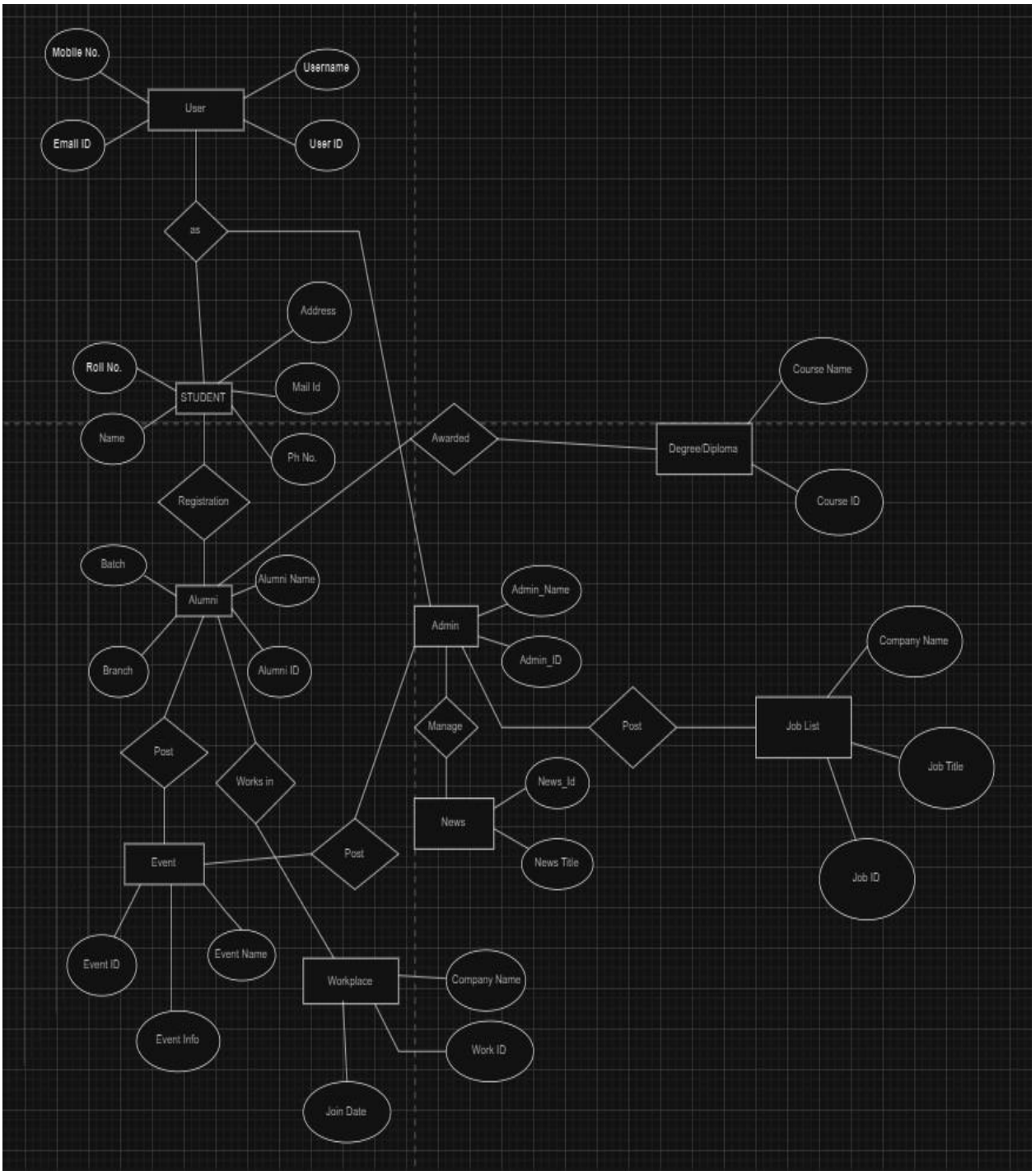


Fig: Scanned UML Diagram for Alumni Management System

ER Diagrams


Mobile No.
Username
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Email ID
User ID
as
Address
Roll No.
STUDENT
Mail Id
Name
Ph No.
Registration
Awarded
Degree/Diploma
Course Name
Course ID
Batch
Alumni
Alumni Name
Branch
Alumni ID
Admin
Admin_Name
Admin_ID
Manage
Post
Job List
Company Name
Job Title
Job ID
Post
Works in
News
News_Id
News Title
Event
Post
Event ID
Event Name
Event Info
Workplace
Company Name
Work ID
Join Date

IMPLEMENTATION DETAILS

i) Hardware Used

- Processor: AMD Ryzen 5
- RAM: 16 GB
- Storage: 512 GB

ii) Software Used

- Database: SQL
- Operating System: Windows 11
- Design and Coding Details: PHP
- Documentation Details: MS Word
- Browser: Brave
- Server: XAMPP

TECHNOLOGIES TO BE USED

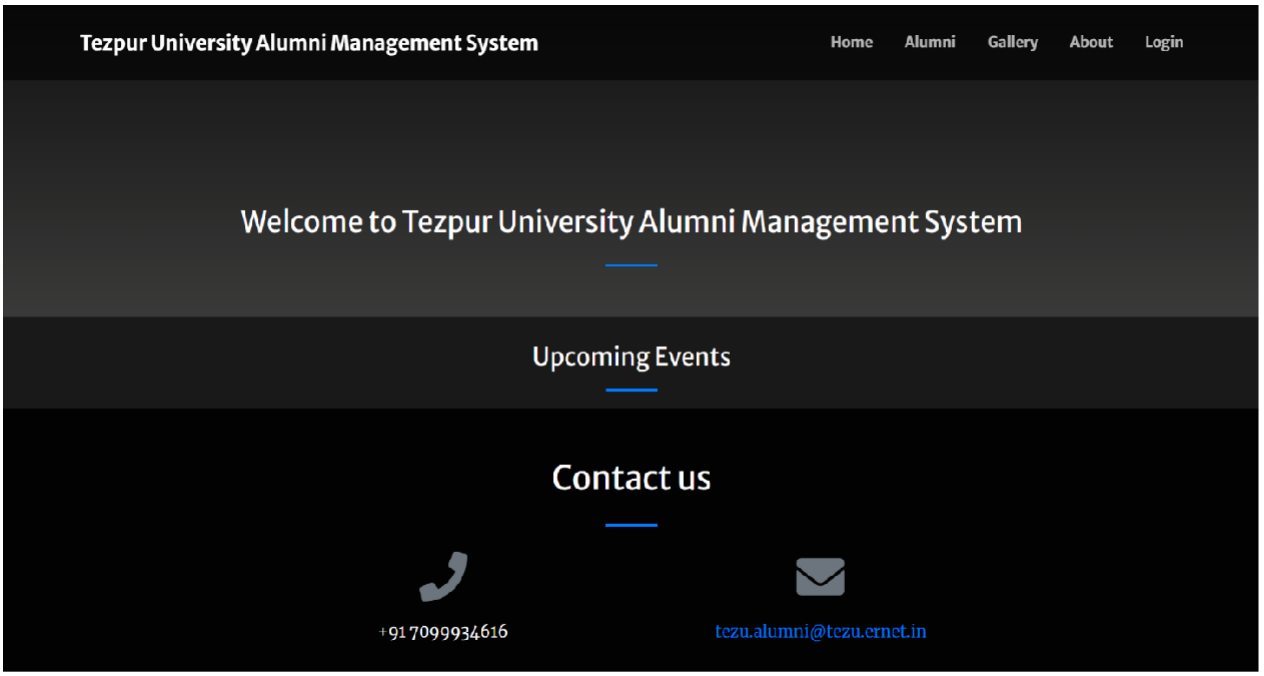
Languages Used

- HTML, CSS: - For front end/UI.
- JQuery: - For adding dynamic effects to elements.
- Bootstrap: - For responsiveness.
- PHP/MySQL: - For backend data processing

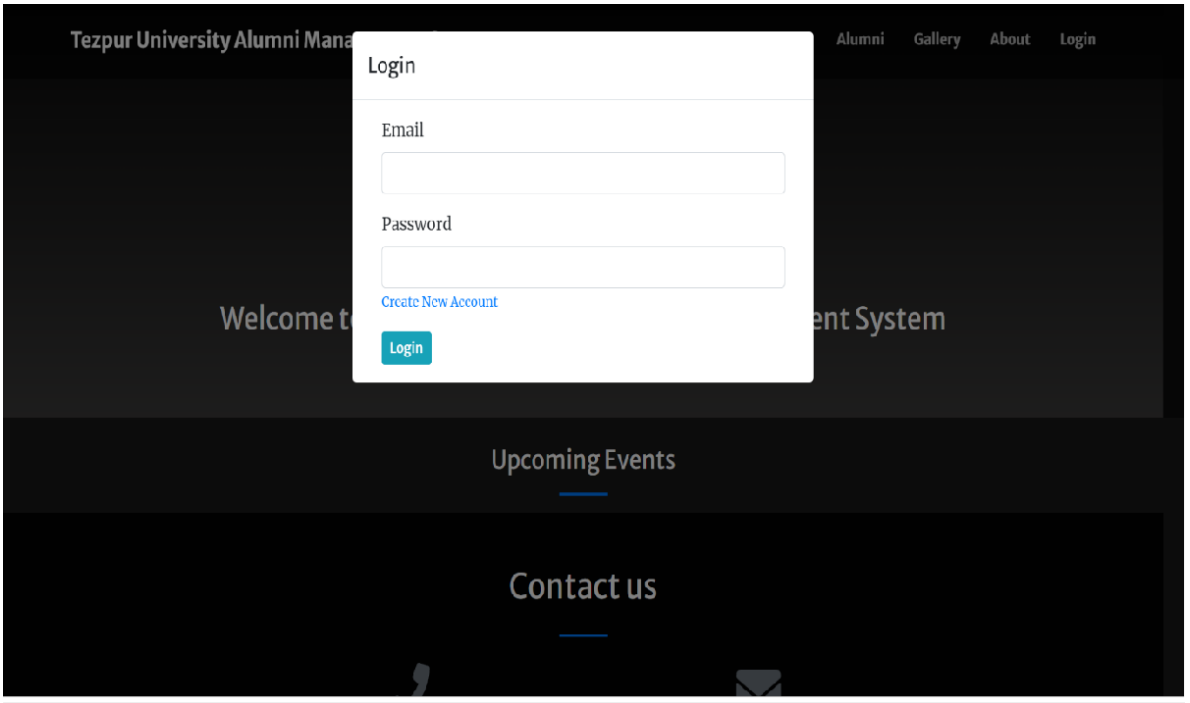
Progress in Testing

Till now I have not completed the following basic testing and I am still working on more features which will take some more but I can assure it will be worth the wait.

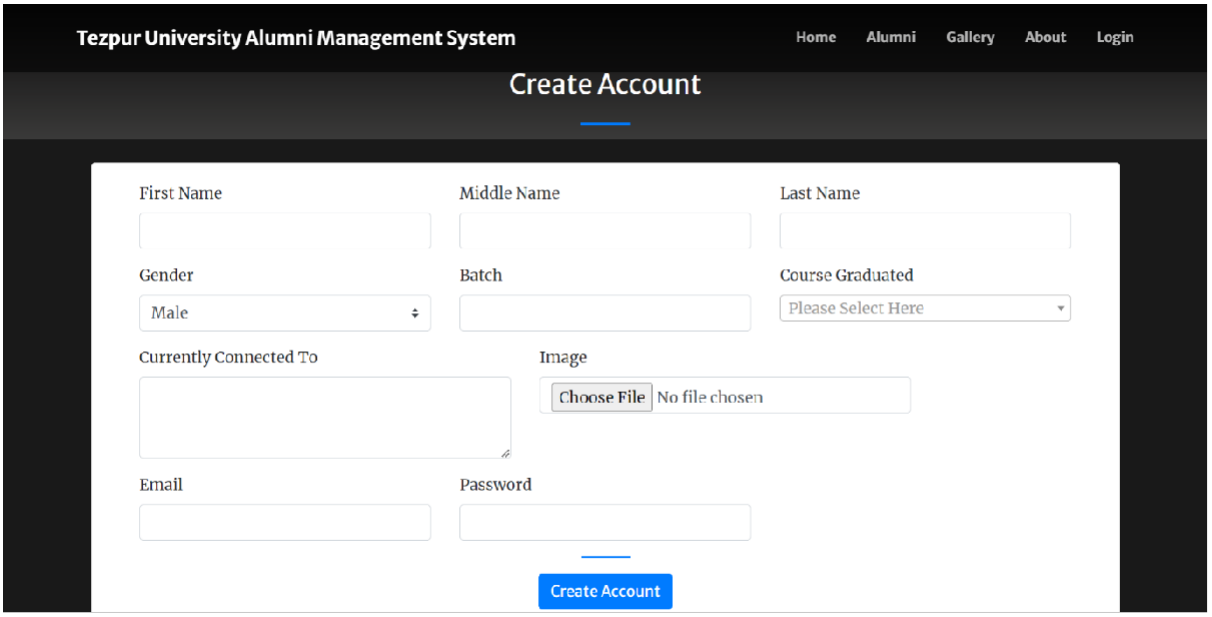
Home Page



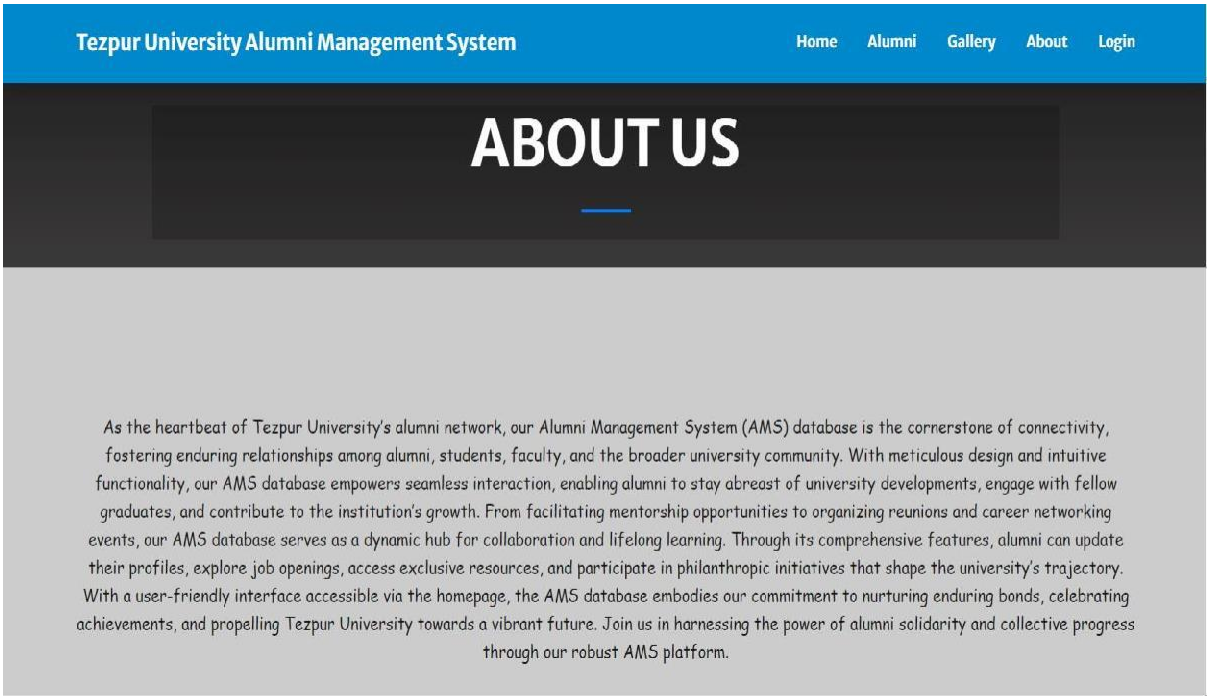
Login Page



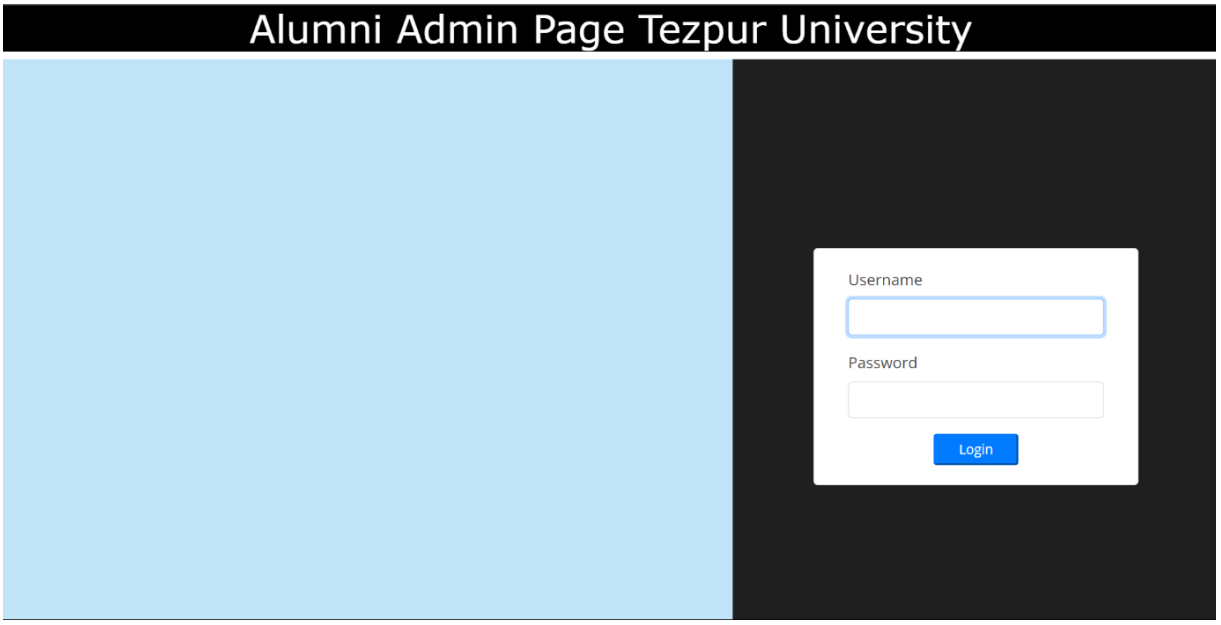
New Account Creation



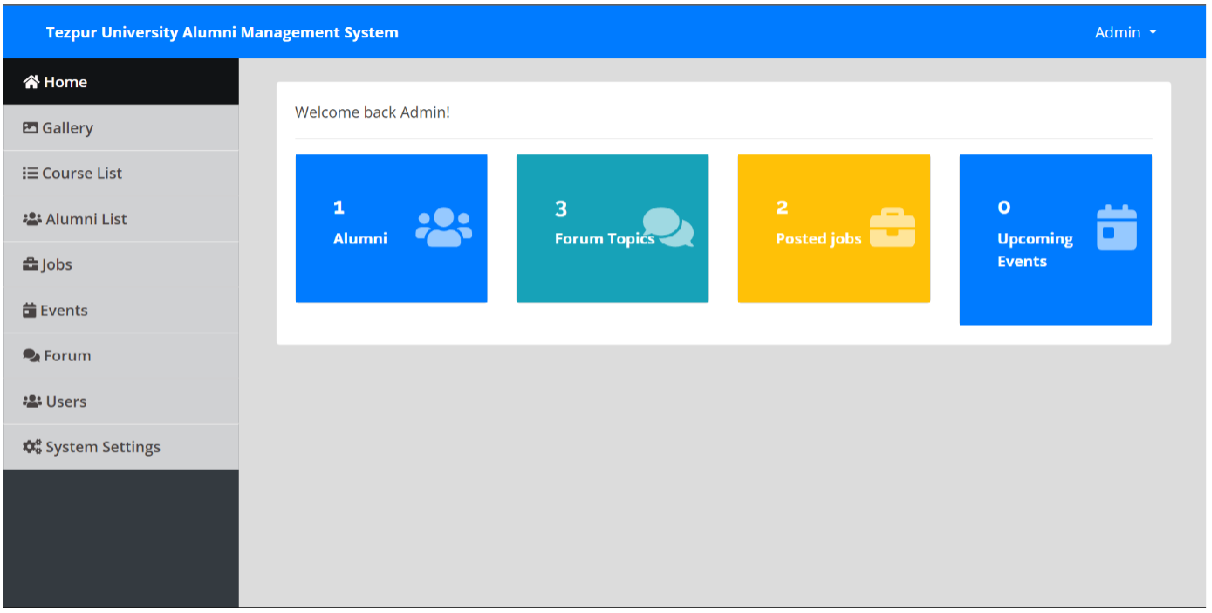
About the Platform



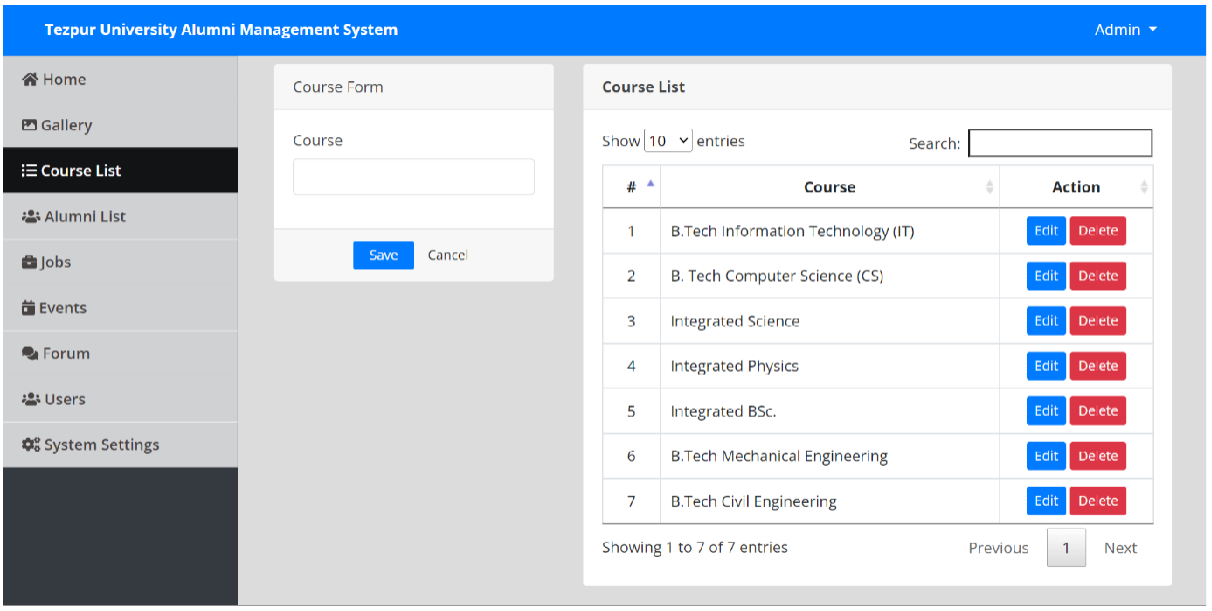
Admin Page



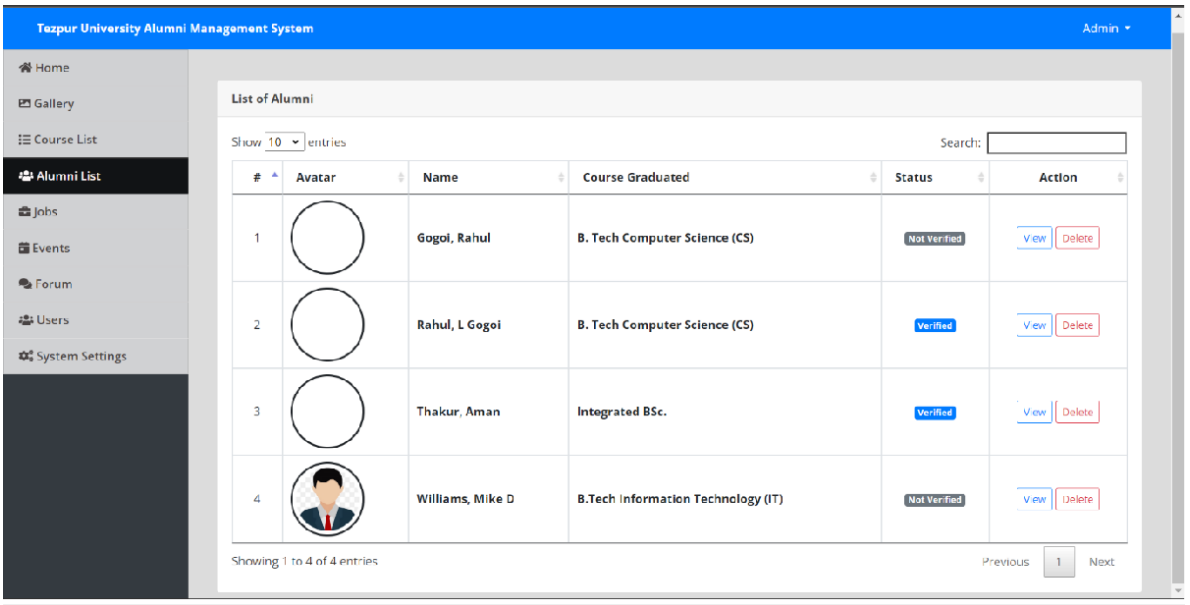
Alumni Admin Dashboard



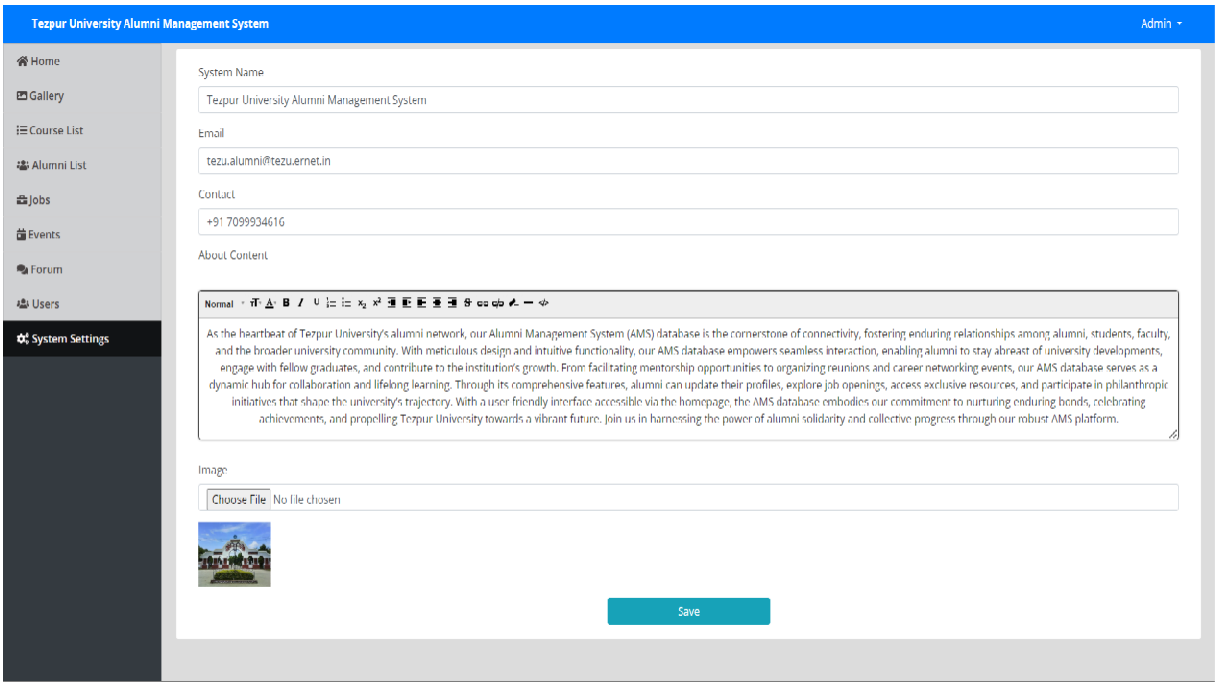
Admin Courses List



Admin Alumni's List

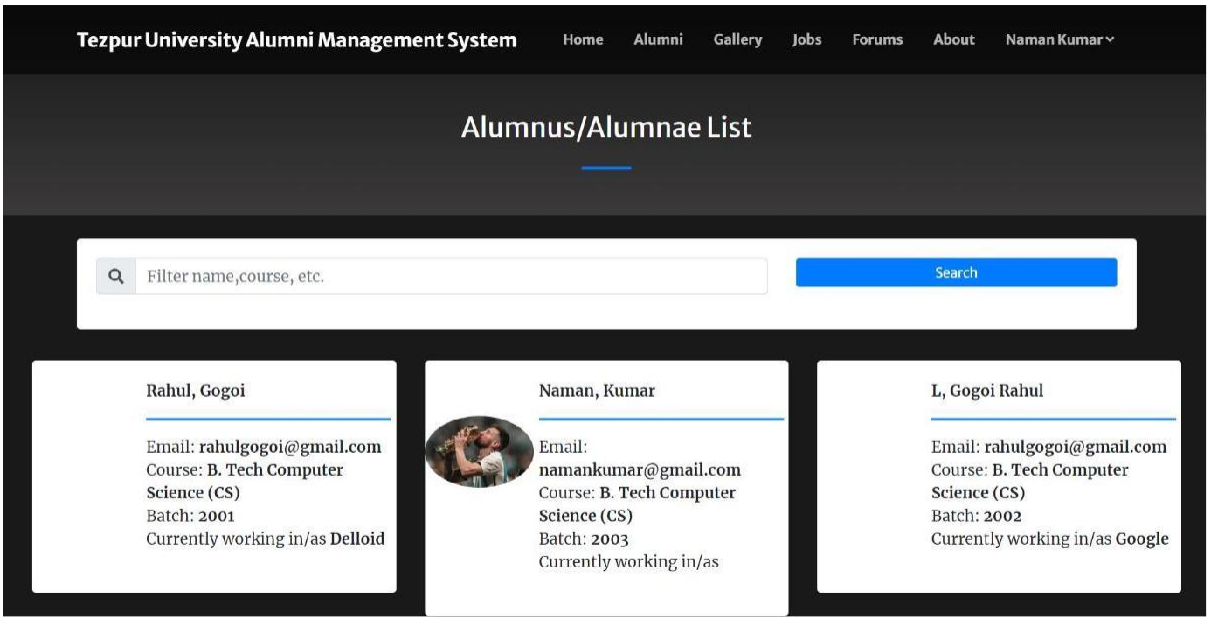


Admin System Settings

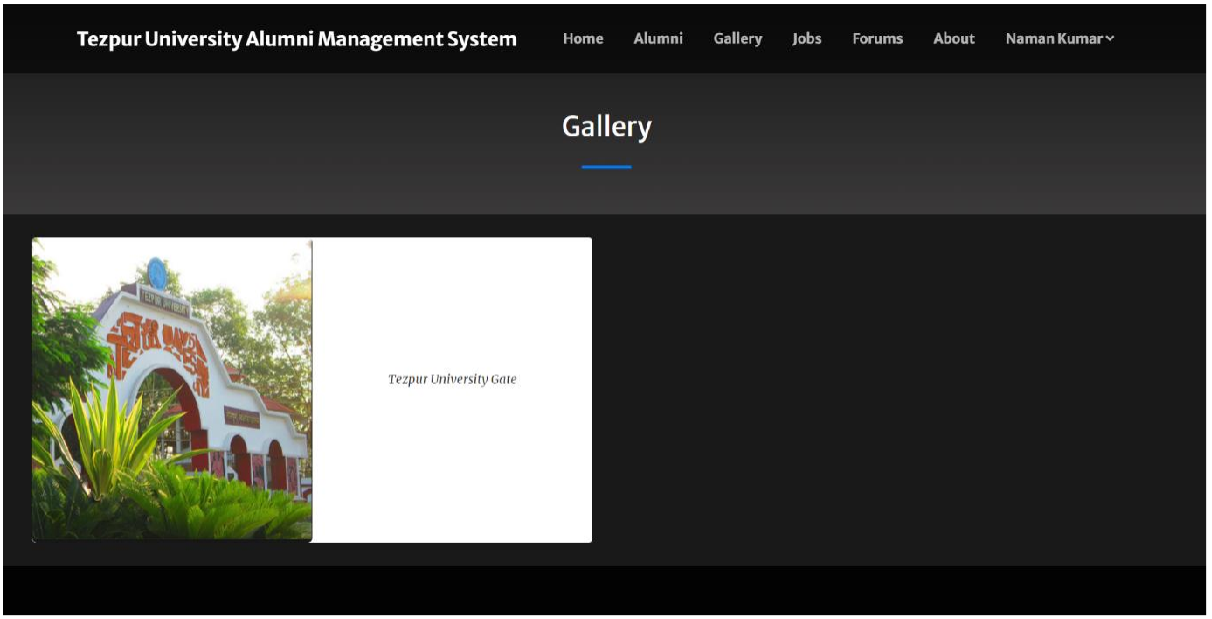


User Dashboard

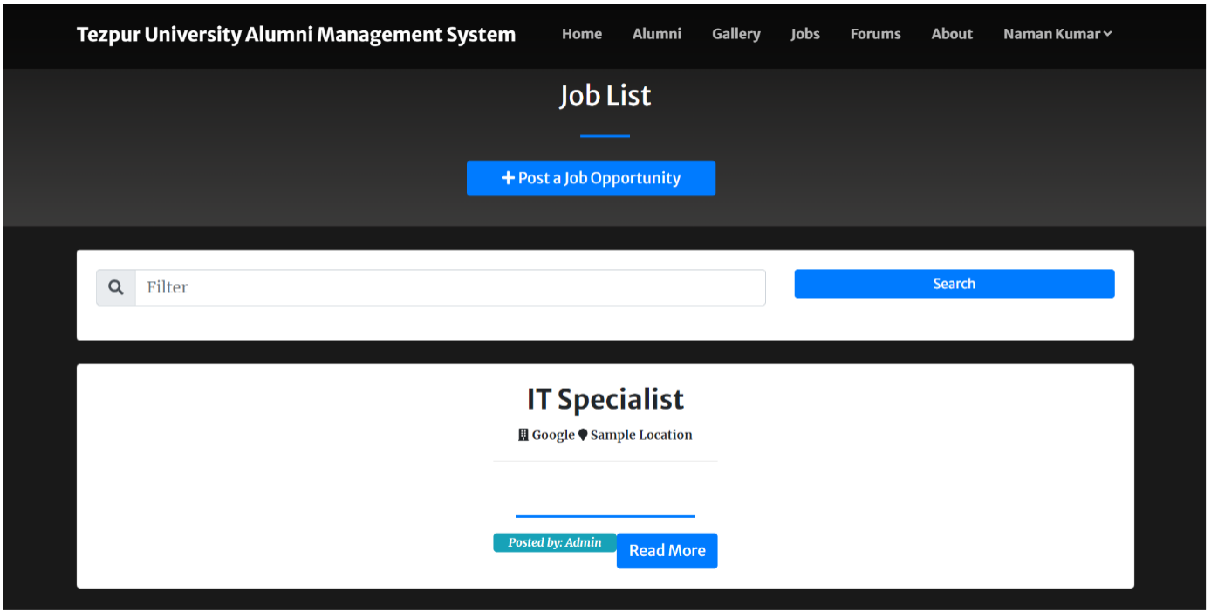
Can look for another alumni information



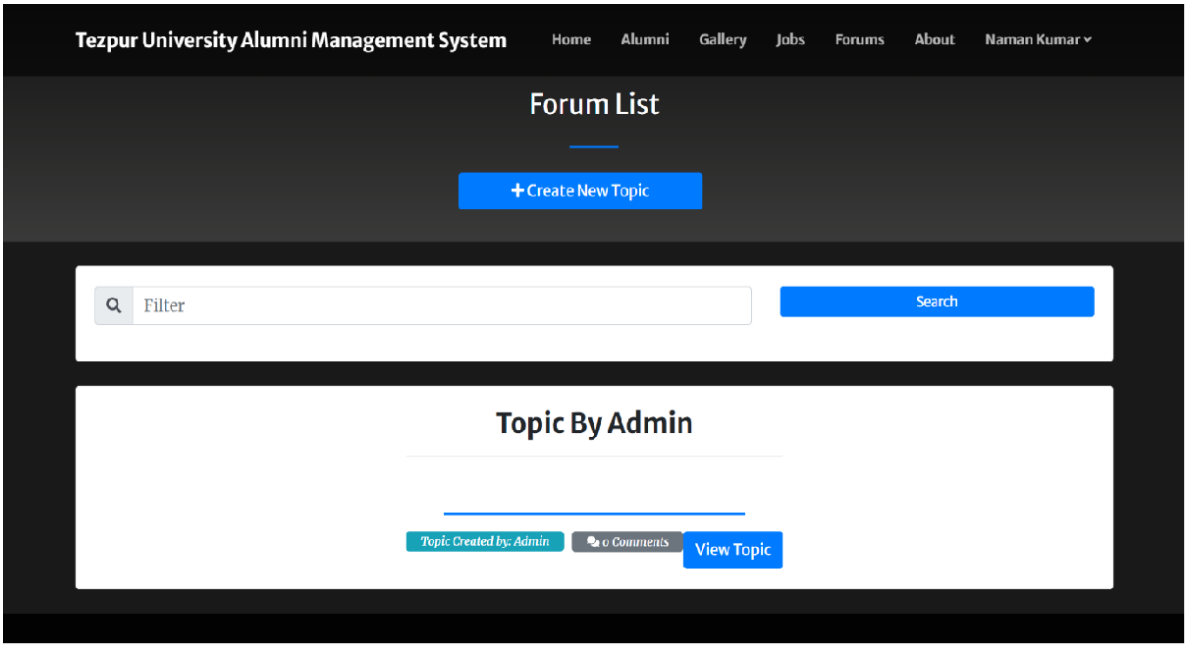
Can see the gallery of the college



Can post a job opportunity and also look for post mode by different people



Can post a Forum



Can see the various upcoming events

CONCLUSION

Future scope:

It would be very useful if the members of the alumni web site which are former student of college could directly contact the alumni officer through the web site. The system that was implemented does not offer this functionality. However, it is easy to find the email address of the alumni officer because it will be placed on the home page of the online community. The contact alumni officer functionality could be easily implemented using asp, which is also used in order to implement the broadcast email functionality that the alumni web site offers. That is why it would be useful to be able to set some information, such as contact details as private or public. This could be done very easy with the use of radio buttons. Right now, alumni website offer only inserting details, later it can be modified to update information. The alumni web site is used to

maintain data of alumni and to provide platform where alumni can see the progress of an institute and also participate in improving institution condition with the help of donation.

Conclusion:

Alumni management system. Effective communication between teachers, old students. Complete automation of all operations centrally stored information with zero redundancy. Alumni web site is one such attempt to bring the passed-out students of the college closer to their juniors. It helps them to connect, get information and guidance for their career.

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